

Major Advanced Topics

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Carrier-Envelope Phase-Dependent Strong-Field Excitation

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Carrier-Envelope Phase-Dependent Strong-Field Excitation

(Phys. Rev. Lett. 128, 173201 (2022))

Carrier Envelope Phase - Surrounding concepts

■ Carrier Envelope Phase

$$\mathbf{F}(t) = F_0 f(t) \sin(\omega t + \varphi) \hat{\mathbf{e}}_x,$$

$$\mathbf{k} = \frac{\omega}{c} \hat{\mathbf{e}}_z$$

■ CEP effects

- Important parameters in the case of few cycles pulses ($n_{oc} < 10$)
- Directly effects most of the relevant observables
- Made evident by a privileged direction in the spectra (anisotropy parameter)
- In the case of Ti:Sapphire, $I \sim 10^{14} \text{ W/cm}^2$, most CEP effects gone by $T \geq 10\tau$

■ Energy level shifts

- Under the field, the fundamental drops by $-\alpha_0 I/4$, with α_0 the polarisability of the atom
- On the other hand, the continuum threshold increases with U_p
- For strong fields, Stark shift also contribute to the energy level shifts

$$U_p = \frac{F_0^2}{4\omega^2}, \quad \gamma = \sqrt{\frac{I_p}{2U_p}}$$

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What to expect from this paper

- Few studies to date have explored the how CEP affects bound-state populations
- CEP can be used to clearly determine when the dynamic changed from Multiphoton- (MP) to tunneling-ionization plus recapture (FTI)
- In this paper, the followings are done
 - Experimental investigation of excitation rates of Ar as a function of CEP in the tunneling regime
 - Showing that bound-state populations dependence on CEP is strong in the tunneling regime
 - Showing at which instant between MP and tunneling regimes, dynamics changes

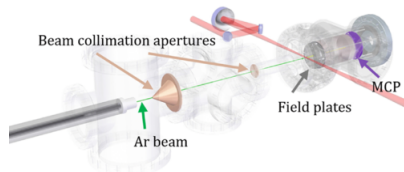
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Experimental setup

- CEP-stabilized laser pulses from Femto Power Pro CE-Phase
 - $I_1 = 2.75 \times 10^{14} \text{ W/cm}^2$ and $I_2 = 2.95 \times 10^{14} \text{ W/cm}^2$
 - 6 fs pulse duration
 - (FWHM) centered at 800 nm,
- Commercial $f - 2f$ interferometer (Menlo APS800)
 - Located close to the interaction region
 - Point is compensating long term drift and measuring the stability of the locking system
- CEP is locked to an absolute (but not absolutely known) value
- Relative shift is controlled by wedge ($1 \text{ mm} \sim 1.23 \text{ rad}$)
- Yields of Ar^* surviving a $0.15 - 0.6 \text{ ms}$ flight time to a microchannel plate detector (MCP) measured
- Time-of-flight used to distinguish yields of Ar^+ from neutral atoms

Chetty *et al.*, Phys. Rev. A 101, 053402 (2020)

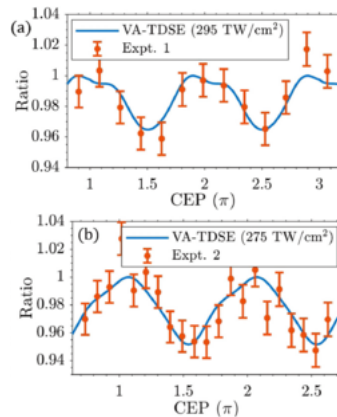


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Experimentally measured and volume averaged TDSE results for the 2 sets

- At intensity $I \leq 2.5 \times 10^{14} \text{ W/cm}^2$, domination of long-term instability of the laser, thus no CEP effect observed
- Ratio of Ar^*/Ar^+ is observed, as experimental systematics are removed
- Similar level of modulation on the order of 5% and a period of π is observed, as required by the symmetry
- Experimentally measured yields are affected by intensity-volume averaging and CEP fluctuations
- Numerical results have been processed to account for those fluctuations
- Agreement between TDSE predictions and experiment shows that numerical results are a good representation of the interaction process

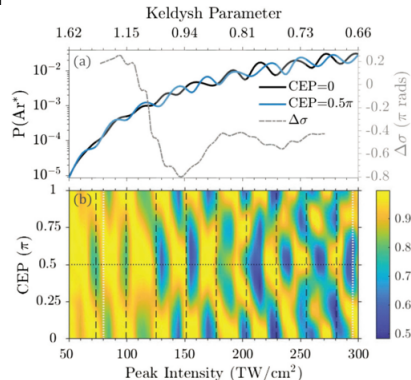


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Total excitation probabilities for TDSE calculations as a function of intensity

- TDSE is solved by splitting the wave function into an inner and outer regions
- Outer wave function is projected into the p -space, and total ioniz. probability is obtained by integration through the space
- Inner-region wave function is projected on the atomic states and summed to get the probability $P(\text{Ar}^*)$
- Fig. 2(a) Comparison of $P(\text{Ar}^*)$ for a cosine and sine pulses.
 - This behaviour changes around $I \sim 1.25 \text{ W/cm}^2$, $\gamma \sim 1$
 - This is further evidenced by the phase shift $\Delta\sigma$ on the figure
 - This indicates the transition from multiphoton- to tunneling-regime
- Fig. 2(b) $P(\text{Ar}^*)$ for each intensity scaled to the the maximum $P(\text{Ar}^*)$ for that intensity
 - Similar patterns repeat at same time as channel-closing
 - Below the 15-photons cc ($\gamma \sim 1$), change is around 20%
 - In the tunneling-regime, change is around 40%



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Details on the numerical calculations-1

- Definition of sine and cosine pulses

$$A(t) = \frac{E_0}{\omega} \sin^2\left(\frac{\pi t}{T}\right) \sin\left(\omega\left(t - \frac{T}{2} + \delta_{\text{CEP}}\right)\right), \quad T = \frac{2\pi N}{\omega}$$

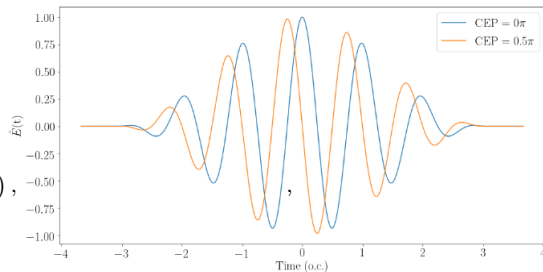
- Comparison of theoretical and experimental yields

- Focal-volume averaging

$$I(r, z) = I_0 \exp\left(-\frac{2r^2}{\omega_0^2}\right), \quad z_a \ll z_R$$

$$r(I') = \omega_0 \sqrt{\frac{1}{2} \ln\left(\frac{I_0}{I'}\right)}, \quad V(I') = \pi z_a r^2(I'),$$

$$Y_{e,i}(\phi, I_{\text{peak}}) = \int_0^{I_{\text{peak}}} P_{e,i}(\phi, I) \frac{dV}{dI} dI,$$



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Details on the numerical calculations-2

- This paper is about numerical calculations that have been proved to be consistent with experiment
- By TDSE calculations, they certainly mean ADK theory and not "numerical resolution of the TDSE"
- Even though this is not explicitly stated in the paper, this is made likely by the cited references

Chetty *et al.*, Phys. Rev. A 101, 053402 (2020)

Tong and Lin, J. Phys. B: At. Mol. Opt. Phys. 38 (2005)

$$W_{\text{TI}}(F) = \frac{C_l^2}{2^{|m|}|m|!} \frac{(2l+1)(l+|m|)!}{2(l-|m|)!} \frac{1}{\kappa^{2Z_c/\kappa-1}} \left(\frac{2\kappa^3}{F} \right)^{2Z_c/\kappa-|m|-1} e^{-2\kappa^3/3F}, \quad \kappa = \sqrt{2I_p}$$

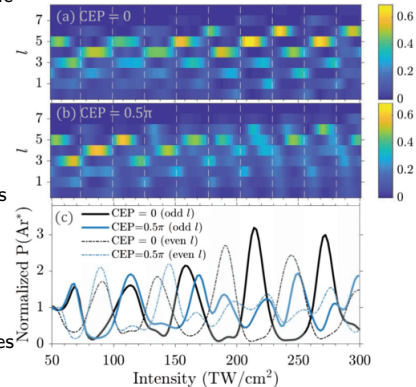
$$P = 1.0 - e^{-\int W_{\text{TI}}[F(t)]dt}$$

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Normalized probabilities for excitation to different l states

- In the MP picture, excitation follows the Ar dipole selection rule
- Authors showed elsewhere that parity effect is observed for CEP-averaged pulses
- It is shown here that this is true for sine and cosine pulses
- Fig. 3(a) and (b) Distribution of l states due to a cosine- and sine- pulses
- Fig. 3(c) Normalized probability for same parity excited states
 - At low intensity, parity effect is observed for both CEPs, consistent with expectation for MP model
 - For cosine pulses, parity is observed at all intensities
 - For sine pulses, parity conservation is lost for $\gamma > 1$
- Venzke *et al.* showed that parity can break down with few cycles pulses
- **Parity depends on CEP for higher intensities**
- Since cosine pulse doesn't break parity, spatial symmetry of the field matters in the tunneling-regime

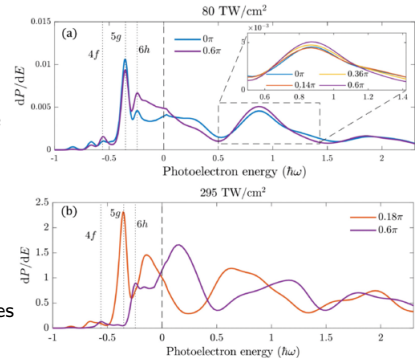


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Normalized excitation and ionization spectra

- Quantum Defect Theory says that photo-excitation and -ionization can be treated the same in the perturbative limit
- Gao and Tong showed that the same holds in the nonperturbative regime
- Fig. 4 shows the normalized photoelectron spectra across the ionization threshold for 80(a) and $295 \times 10^{14} \text{ W/cm}^2$ (b)
- CEP pertaining to min and max excitation yields are shown
- (a) Lower intensity is dominated by the 13-photon resonance (5g state)
 - With CEP from 0 to 0.6π , 15% change in modulation
 - Change is less significant for bound states of lower energies
- (b) For higher intensity, drastic change can be seen all along
 - Both structure and positions of the peaks are modified with the CEP
 - Populations of individual states is extremely CEP-sensitive



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Closure

- Bound-states population dependence on the CEP of the laser field has been shown
 - Hopefully, this finding will encourage the use of other laser techniques (like 2 colors fields) to control bound-state populations
 - CEP should also allow a better exploration of the boundary between multiphoton and tunneling regime
 - CEP causes change in modulation and structure, but also the relative phase between the peaks in the tunneling regime
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- Those phenomena warrant further investigation with alternative experimental methods